

Lab04_Franco

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Questão 1

```
library(readxl)
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0
—
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

Questão 2

```
fname <- "WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

Questão 3

```
countries <- read_excel(fname, sheet = "Country")
```

Questão 4

```
countries <- countries %>%
  filter(`Country Code` %in% c("BRA", "ARG", "USA", "CHN", "IND"))
countries
```

```
# A tibble: 5 × 31
  `Country Code` `Short Name` `Table Name` `Long Name`      `2-alpha
code`
  <chr>          <chr>        <chr>        <chr>
<chr>
1 ARG           Argentina   Argentina   Argentine Republic
AR
2 BRA           Brazil      Brazil      Federative Republic...
BR
3 CHN           China       China       People's Republic o...
CN
4 IND           India       India       Republic of India
IN
5 USA           United States United States United States of Am...
US
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   `Region` <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor`
<lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Questão 5

O arquivo WDI_Data.csv original possui quase 400 mil linhas (referentes a todos os países do mundo), o que excedia a memória disponível no ambiente do Posit Cloud e causava falhas ao tentar processá-lo. Por esse motivo, o arquivo foi pré-filtrado via terminal (comando grep), mantendo apenas as linhas referentes aos 5 países selecionados na Questão 4, antes da leitura no R.

```
WDI <- read_csv2("WDI_Data_filtrado.csv")
```

```
i Using "','" as decimal and "'.'" as grouping mark. Use `read_delim()` for
more control.
```

```
Rows: 7490 Columns: 70
— Column specification
```

```
Delimiter: ";"
chr (4): Country Name, Country Code, Indicator Name, Indicator Code
dbl (66): 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968, 1969,
1970, ...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Questão 6

```
WDI <- WDI %>%  
  select(`Country Code`, `Indicator Code`, `1960`:`2025`)
```

Questão 7

```
glimpse(WDI)
```

```
Rows: 7,490  
Columns: 68  
$ `Country Code` <chr> "ARG", "ARG", "ARG", "ARG", "ARG", "ARG", "ARG",  
"ARG...  
$ `Indicator Code` <chr> "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.RU.ZS",  
"EG.CFT.ACCS.U...  
$ `1960` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1961` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1962` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1963` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1964` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1965` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1966` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1967` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1968` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1969` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1970` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1971` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1972` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...  
$ `1973` <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,  
N...
```

\$ `1974` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1975` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1976` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1977` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1978` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1979` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1980` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1981` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1982` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1983` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1984` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1985` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1986` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1987` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1988` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1989` N...	<dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
\$ `1990` 9.413902e+01,...	<dbl> NA, NA, NA, 9.215480e+01, 7.889459e+01,
\$ `1991` 9.441312e+01,...	<dbl> NA, NA, NA, 9.249219e+01, 7.925427e+01,
\$ `1992` 9.468670e+01,...	<dbl> NA, NA, NA, 9.282945e+01, 7.977864e+01,
\$ `1993` 9.495712e+01,...	<dbl> NA, NA, NA, 9.316605e+01, 8.033370e+01,
\$ `1994` 9.522123e+01,...	<dbl> NA, NA, NA, 9.350127e+01, 8.093589e+01,
\$ `1995` 9.547585e+01,...	<dbl> NA, NA, NA, 9.383447e+01, 8.160809e+01,
\$ `1996` 9.571784e+01,...	<dbl> NA, NA, NA, 9.416494e+01, 8.236988e+01,
\$ `1997` 9.594404e+01,...	<dbl> NA, NA, NA, 9.449201e+01, 8.324706e+01,

\$ `1998` 9.615127e+01,...	<dbl> NA, NA, NA, 9.481500e+01, 8.426384e+01,
\$ `1999` 9.633807e+01,...	<dbl> NA, NA, NA, 9.513316e+01, 8.543186e+01,
\$ `2000` 9.570000e+0...	<dbl> 9.580000e+01, 7.760000e+01, 9.830000e+01,
\$ `2001` 9.550000e+0...	<dbl> 9.630000e+01, 8.010000e+01, 9.860000e+01,
\$ `2002` 9.610000e+0...	<dbl> 9.690000e+01, 8.220000e+01, 9.870000e+01,
\$ `2003` 9.630000e+0...	<dbl> 9.720000e+01, 8.430000e+01, 9.890000e+01,
\$ `2004` 9.650000e+0...	<dbl> 9.760000e+01, 8.620000e+01, 9.910000e+01,
\$ `2005` 9.670000e+0...	<dbl> 9.800000e+01, 8.770000e+01, 9.920000e+01,
\$ `2006` 9.700000e+0...	<dbl> 9.830000e+01, 8.910000e+01, 9.940000e+01,
\$ `2007` 9.720000e+0...	<dbl> 9.860000e+01, 9.060000e+01, 9.950000e+01,
\$ `2008` 9.750000e+0...	<dbl> 9.880000e+01, 9.200000e+01, 9.960000e+01,
\$ `2009` 9.780000e+0...	<dbl> 9.905000e+01, 9.280000e+01, 9.970000e+01,
\$ `2010` 9.880000e+0...	<dbl> 9.920000e+01, 9.380000e+01, 9.970000e+01,
\$ `2011` 9.900000e+0...	<dbl> 9.940000e+01, 9.440000e+01, 9.970000e+01,
\$ `2012` 9.910000e+0...	<dbl> 9.950000e+01, 9.530000e+01, 9.980000e+01,
\$ `2013` 9.930000e+0...	<dbl> 9.960000e+01, 9.570000e+01, 9.980000e+01,
\$ `2014` 1.000000e+0...	<dbl> 9.970000e+01, 9.630000e+01, 9.990000e+01,
\$ `2015` 9.970000e+0...	<dbl> 9.980000e+01, 9.650000e+01, 9.990000e+01,
\$ `2016` 9.990000e+0...	<dbl> 9.980000e+01, 9.690000e+01, 9.990000e+01,
\$ `2017` 1.000000e+0...	<dbl> 9.980000e+01, 9.730000e+01, 9.990000e+01,
\$ `2018` 1.000000e+0...	<dbl> 9.990000e+01, 9.750000e+01, 9.990000e+01,
\$ `2019` 1.000000e+0...	<dbl> 9.990000e+01, 9.780000e+01, 9.990000e+01,
\$ `2020` 1.000000e+0...	<dbl> 9.990000e+01, 9.780000e+01, 9.990000e+01,
\$ `2021` 1.000000e+0...	<dbl> 9.990000e+01, 9.810000e+01, 9.990000e+01,

```
$ `2022`      <dbl> 99.90000, 98.20000, 99.90000, 100.00000, 100.00000,
1...
$ `2023`      <dbl> 99.9, 98.2, 99.9, 100.0, 100.0, 100.0, NA, NA, NA,
NA...
$ `2024`      <dbl> NA, NA, NA, 98.40000, 79.80000, 100.00000, 81.74424,
...
$ `2025`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
N...
```

Dá pra ver que cada ano virou uma coluna, então não está tidy (o certo seria ter uma coluna só pra ano). Usei `pivot_longer` pra resolver isso.

```
WDI <- WDI %>%
  pivot_longer(cols = `1960`:`2025`, names_to = "YEAR", values_to = "value")

WDI
```

```
# A tibble: 494,340 × 4
  `Country Code` `Indicator Code` YEAR value
  <chr>          <chr>          <chr> <dbl>
1 ARG           EG.CFT.ACCS.ZS  1960    NA
2 ARG           EG.CFT.ACCS.ZS  1961    NA
3 ARG           EG.CFT.ACCS.ZS  1962    NA
4 ARG           EG.CFT.ACCS.ZS  1963    NA
5 ARG           EG.CFT.ACCS.ZS  1964    NA
6 ARG           EG.CFT.ACCS.ZS  1965    NA
7 ARG           EG.CFT.ACCS.ZS  1966    NA
8 ARG           EG.CFT.ACCS.ZS  1967    NA
9 ARG           EG.CFT.ACCS.ZS  1968    NA
10 ARG          EG.CFT.ACCS.ZS  1969    NA
# i 494,330 more rows
```

Questão 8

```
class(WDI$YEAR)
```

```
[1] "character"
```

É character mesmo, então é preciso converter pra integer.

```
WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))

class(WDI$YEAR)
```

```
[1] "integer"
```

Questão 9

A coluna Short Name está presente apenas no objeto countries, não em WDI. Portanto, não é suficiente apenas filtrar as linhas: é necessário trazer essa coluna para dentro de WDI por meio de um join. Como o objetivo é adicionar uma nova coluna (e não somente filtrar linhas), trata-se de um mutating join (inner_join), e não de um filtering join.

O filtro dos países já havia sido aplicado na Q6, no momento da leitura dos dados, pois sem esse filtro o pivot_longer da Q7 ultrapassava a capacidade de memória disponível (o conjunto de dados original possui quase 400 mil linhas, referentes a todos os países do mundo).

Antes do join, foi aplicado um pivot_wider para reorganizar os indicadores em colunas, permitindo selecionar diretamente a variável SP.DYN.CBRT.IN.

```
WDI <- WDI %>%
  pivot_wider(names_from = `Indicator Code`, values_from = value) %>%
  inner_join(countries %>% select(`Country Code`, `Short Name`), by = "Country
Code") %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
```

WDI

```
# A tibble: 330 × 4
  `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
  <chr>          <chr>        <int>    <dbl>
1 ARG           Argentina    1960     24.2
2 ARG           Argentina    1961     23.9
3 ARG           Argentina    1962     23.8
4 ARG           Argentina    1963     23.5
5 ARG           Argentina    1964     23.2
6 ARG           Argentina    1965     22.6
7 ARG           Argentina    1966     22.3
8 ARG           Argentina    1967     22.4
9 ARG           Argentina    1968     22.7
10 ARG          Argentina    1969     23.0
# i 320 more rows
```

Questão 10

```
WDI <- WDI %>%
  arrange(`Country Code`, YEAR)
```

Questão 11

```
WDI %>%
  filter(!is.na(SP.DYN.CBRT.IN)) %>%
  ggplot(aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line() +
  labs(x = "Ano", y = "Taxa de natalidade (SP.DYN.CBRT.IN)", color = "País",
       title = "Taxa de natalidade ao longo do tempo")
```

